🧩 M002 Extension 4 — Complete the Maze

Topic: Add Two AND Conditions · **Course:** Maze Madness · **Level:** Extension (Intermediate) · **Time:** about 45 minutes

The agent follows a redstone trail with **four rules**, but the solver is **not finished**. At two junctions two signals fire at once and it stops — add two **AND** rules, then drive it to the goal.

Chat: `l` turn left, `r` turn right, `run` start solver, `rl` bring agent back.

1 · The Four Rules 🎛️

Each rule checks **one** signal and does **one** action.

```
if agent.detect(REDSTONE, LEFT):
    agent.turn(RIGHT_TURN)
elif agent.detect(REDSTONE, RIGHT):
    agent.turn(LEFT_TURN)
elif agent.detect(REDSTONE, DOWN):
    agent.move(UP, 1)
elif agent.detect(REDSTONE, UP):
    agent.move(DOWN, 1)
```

Fill the table from the code. You will reuse these actions.

Signal	Action
redstone **left**	
redstone **right**	
redstone **below** (down)	
redstone **above** (up)	

2 · Find the Two Stuck Junctions 🔍

Open the **M002 Complete the Maze** world and type `run`. Watch where the agent stops, then look around it.

Tick the signals on at each stuck junction.

	left	right	below	above
Junction 1				
Junction 2				

Only the first matching `elif` fires. Why does that leave the second signal unused?

3 · AND means BOTH 🧠

An **AND** is true **only when both** halves are true.

```
if agent.detect(REDSTONE, LEFT) and agent.detect(REDSTONE, DOWN):
    ...
```

Example: left on but NOT below. Is the whole condition true?

Look at Junction 1 in your table. Write the AND condition that is true only there.

4 · Add the Two New Rules 🛠️

Add **two** `elif` **branches**, one per stuck junction, using the signals **you** found. Each branch does the action for **each** signal.

Junction 1 — fill the blanks:

```
elif agent.detect(REDSTONE, ______) and agent.detect(REDSTONE, ______):
    agent.________(___________)
    agent.________(___________)
```

Junction 2 — write the whole branch yourself:

Put both new rules *before* the four single-signal rules. Why?

5 · Spot the Bug 🐛

This branch should fire **only when both** signals are on. It fires when **either** one is on, so it triggers at wrong spots.

```
elif agent.detect(REDSTONE, LEFT) or agent.detect(REDSTONE, DOWN):
    agent.turn(RIGHT_TURN)
    agent.move(UP, 1)
```

Write the fixed code:

Why was it wrong?

6 · Show Your Work 📸 🎥

Type `run`. Goal: start to the **very end**, past both stuck junctions — fix your rules and run again until it reaches the goal.

Record **one video** — one take, no stopping (a phone is fine). Show these in order:

1 · Your code. Scroll through it. Say what each part does.

2 · Your build. Point the camera. Name the parts.

Fill the blanks:

> Today I built _____.
>
> I built it using this code: _____.
>
> In this code I used _____.
>
> The hardest part was _____.
>
> That part was hard because _____.
>
> The most fun part was _____.
>
> Something new I learned was _____.

Write your lines here, then say them in your video.

Submit ✅

Send this worksheet + your video to teacher on KakaoTalk.